

The total possible points are 60:

P1(30): being able to retrieve the data from NIST worth 10 points; having all equations right worth 10 points; plots of Shomate equation and Murphy worth 5 points; the right solution of the problem worth 5 points.

P2(30): being able to derive  $f$  and  $\phi$  in terms of  $T$  worth 8 points; being able to derive the equation of  $\psi$  worth 8 points; deriving  $f^{\sim}P$  and  $T^{\sim}P$  relations correctly worth 8 points; having the right plots worth 6 points.